CSS 2201- DBS Theory-

MidTerm Exam -QP-Solutions

Q.No	Question	Marks	CO	BT
Q1-10	**Q1.** A transaction updates several tables and then crashes before commit. Which component ensures that none of the partial changes become visible after recovery? A. Storage manager B. Buffer manager C. Transaction manager D. Authorization manager **Answer: C** **Q2.** Which relational algebra expression retrieves instructors in the Physics department? A. π name (instructor × department) B. σ dept_name = 'Physics' (instructor) C. instructor ⋈ department D. instructor − department **Answer: B** **Q3.** Which SQL query correctly returns instructors whose salary is **greater than at least one salary** in the Biology department, even when the subquery produces duplicates? A. salary > ANY (SELECT salary FROM instructor WHERE dept_name='Biology') B. salary > ALL (SELECT salary FROM instructor WHERE dept_name='Biology') C. salary IN (SELECT salary FROM instructor WHERE dept_name='Biology') D. salary NOT IN (SELECT salary FROM instructor WHERE dept_name='Biology') **Answer: A** **Q4.** Which join operation can unintentionally filter out required results if two relations share an attribute name that is unrelated semantically? A. Inner join using ON B. Natural join C. Left outer join D. Right outer join **Answer: B** **Q5.**Which SQL command removes a table completely from the database? A. DELETE B. REMOVE C. DROP D. TRUNCATE **Answer: C** **Q6.** Which operator checks if a value matches any value in a list?	10*0.5 =5	1,2	2

	A. ANY B. BETWEEN C. IN D. EXISTS **Answer:** C **Q7.** Which wildcard represents a single character in a LIKE condition? A. % B. _ C. * D. ? **Answer:** B **Q8.** Which condition is used to check for NULL values? A. = NULL B. IS NULL C. NULL = D. EQUAL NULL **Answer: B** **Q9.** Which statement is TRUE about GROUP BY? A. It can be used without SELECT B. All selected columns must be aggregated C. Non-aggregated columns must appear in GROUP BY D. It works only with WHERE Answer: C **Q10.**Which operator is commonly used with subqueries returning multiple rows? A. = B. LIKE C. IN D. IS **Answer: C**			
Q11.	a) Assume a table: **Employee(id, name, salary)** Write a query to find employees earning more than the minimum salary but less than the maximum salary. b) A hospital database contains: **Patient**(PatientID, Name, Diagnosis, Treatment, BillingAmount). Doctors should access all attributes, while reception staff should access only PatientID, Name, and BillingAmount. Give the corresponding SQL statement(s) to create a view named reception_view that enforces the above access restriction. Discuss how views enforce security and simplify access control. **Answer: a) -> 2M + b)→2M** a) SELECT name FROM Employee WHERE salary > (SELECT MIN(salary) FROM Employee) AND salary < (SELECT MAX(salary) FROM Employee); b) CREATE VIEW Reception_View AS	**4**	**2**	**3**

	SELECT PatientID, Name, BillingAmount FROM Patient; Restrict access to sensitive attributes:**Views hide confidential columns** such as Diagnosis and Treatment. Prevent unauthorized direct access:**Users are granted access only to the view**, not the base table.			
Q12.	Consider the Parts database given below: Parts(PID, Name, Color) Supplier(SID, Name, City) Catalog(PID, SID, Price, Destination) Provide the relational algebra expressions for the following: a) Identify the suppliers who have supplied at least one part to a destination that is the same as their own city. b) Find the PIDs of parts that are priced more than all the red colored parts. **Answer:** a) $\boxed{\pi_{SID,Name,City}\left(\sigma_{City=Destination}\left(Supplier \bowtie_{Supplier.SID=Catalog.SID} Catalog\right)\right)}$ b) Red_Prices ← $\pi_{PID,Price}(\sigma_{Color='Red'}(Parts) \bowtie Catalog)$ R1← $\pi_{P.Price,P.PID}(\sigma_{P.Price<Red_{Prices}.Price}(Red_Prices \times \rho_P(Parts \bowtie Catalog))$ Result ← $\pi_{PID}(Parts) - \pi_{PID}(\mathrm{R1})$ Scheme for evaluation: A) 1 Mark – 0.5 for join, 0.5 for select and project B) 2 Marks – 0.5 for Red_Prices, 1 for R1, 0.5 for Result.	**3**	**1**	**3**
Q13.	A media organization intends to develop a structured database system to efficiently manage information related to movies, production companies, actors, directors, and movie categories. Each movie is uniquely identified by the combination of Title and Year of Release and includes attributes such as Length (in minutes) and Language. A movie may also contain zero or more quotes, which must be stored in the system. Each production company is uniquely identified by its Name and maintains details such as Address. A production company can produce one or more movies,	**3**	**3**	**4**

while every movie must be produced by exactly one production company. Actors are uniquely identified by the combination of Name and Date of Birth. An actor may perform in one or more movies, and for every movie in which an actor appears, the specific role played by the actor must be recorded. Similarly, directors are uniquely identified by Name and Date of Birth. A director can direct one or more movies, and a movie may be directed by one or more directors. Furthermore, each movie must be classified under exactly one category (such as Horror, Action, Drama, etc.), and each category can include multiple movies.

Analyze the above requirements and design a comprehensive ER diagram. Clearly identify entities, attributes (including composite or multivalued attributes, if any), primary keys, relationship sets, cardinalities, and participation constraints.

Answer:

ER Diagram

- **Entities, Primary Keys (1mark)**

Movie

- Primary Key: (Title, Year_of_Release)
- Length
- Language
- Quotes

Production_Company

- Primary Key: Name
- Address

Actor

- Primary Key: (Name, Date_of_Birth)

Director

- Primary Key: (Name, Date_of_Birth)

Category

- Primary Key: Category_Name

Special Attribute Types: (0.5 mark)

Multivalued Attribute:

- **Quotes (of Movie)**

Relationship Attribute:

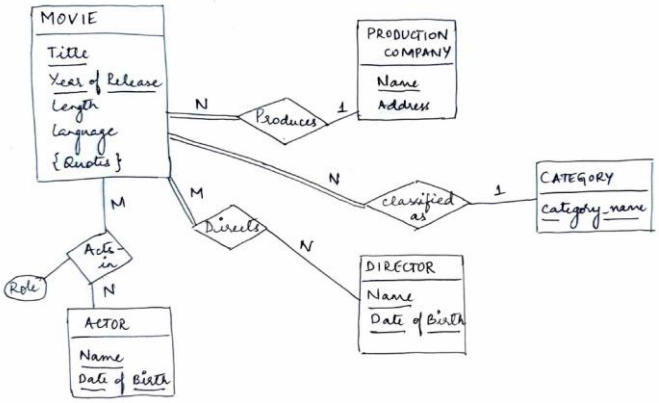
- Role (in Acts_In relationship)

2. Relationship Sets with Cardinalities and participation constraints (1 mark)

Relationship attribute: (0.5 mark)

Produces (Between: Production_Company — Movie)

- **Cardinality Ratio: Production_Company 1: N Movie**

	• **Participation Constraint:** • **Movie:** Total participation (every movie must be produced by exactly one company) • **Production_Company:** Partial participation **Acts_In(Between: Actor — Movie)** • **Cardinality Ratio: M : N** • **Relationship Attribute: Role** (role played by actor in a movie) • **Participation Constraint:** ❖ **Actor:** Partial ❖ **Movie:** Partial **Directs (Between: Director — Movie)** • **Cardinality Ratio: M : N** • **Participation Constraint:** ❖ **Movie**: Total participation (every movie must have at least one director) ❖ **Director**: Partial **Classified_As (Between: Category — Movie)** • **Category (1) —— (N) Movie** • **Cardinality Ratio:** Category **1 : N** Movie • **Participation Constraint:** ❖ **Movie:** Total participation (each movie belongs to exactly one category) ❖ **Category:** Partial 			
Q14.	Reduce the given ER diagram to the equivalent relational schemas by applying the standard ER-to-Relational mapping procedure. Clearly specify the attributes, Primary Key (PK) and Foreign Key (FK) constraints for each relation.	**3**	**3**	**4**

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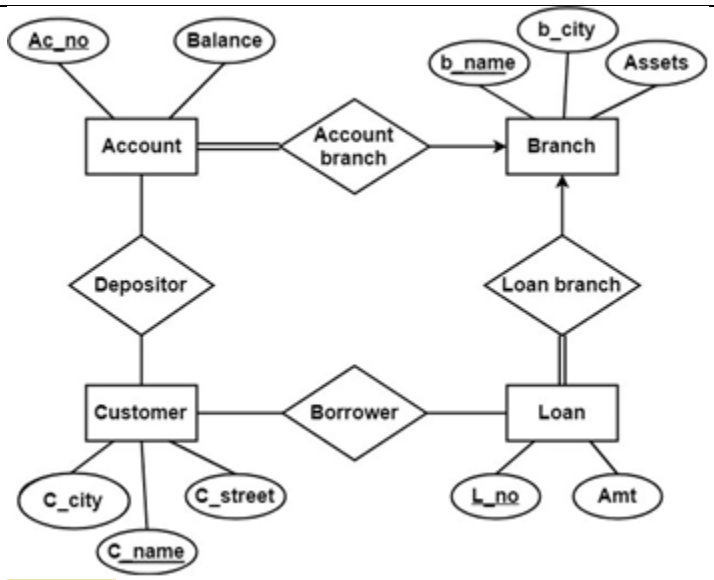

Answer:
Applying the rules minimum 6 tables will be required-
• Account (<u>Ac_no,</u> Balance, b_name)
PK: Ac_no
FK: b_name → Branch(b_name)
• Branch (<u>b_name</u>, b_city, Assets)
PK: b_name
• Loan (<u>L_no</u>, Amt, b_name)
PK: L_no
FK: b_name → Branch(b_name)
• Customer (<u>C_name</u>, C_street, C_city)
PK: C_name
• Depositor (<u>C_name</u>, <u>Ac_no</u>)
PK: (C_name, Ac_no)
FK: C_name → Customer(C_name)
FK: Ac_no → Account(Ac_no)
• Borrower (<u>C_name</u>, L_no)
PK: (C_name, L_no)
FK: C_name → Customer(C_name)
FK: L_no → Loan(L_no)
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<td>Q15.</td>
<td>Consider the following relational schemas:
employee(emp_id, name, dept_id)
project(project_id, project_name)
works_on(emp_id, project_id)

a) Write a sql query using set operation to find the emp_id of employees who work on project 101 or belongs to department 5.(1mark)</td>
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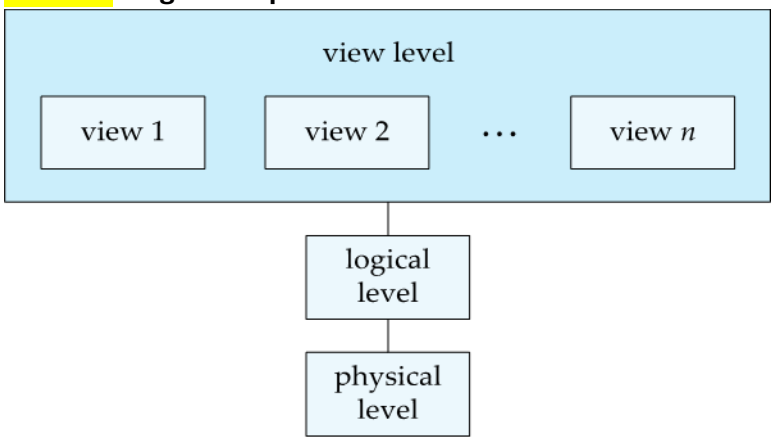
	b) Write an SQL query using the set Operation to find employee IDs who work on at least one project that emp_id 301 works on. (2marks)
	Answer :
	a)
	(SELECT emp_id
	FROM works_on
	WHERE project_id = 101) (0.5 mark)
	
	UNION
	
	(SELECT emp_id
	FROM employee (0.5 mark)
	WHERE dept_id = 5);
	b)
	SELECT DISTINCT W.emp_id
	FROM works_on W // If at least one common
	//project exists, that employee
	// ID is selected. (1 mark)
	WHERE EXISTS
	(
	SELECT project_id
	FROM works_on
	WHERE emp_id = W.emp_id // Find common projects
	//using INTERSECT
	//between
	// current employee and
	//employee 101 (0.5mark)
	
	INTERSECT
	
	SELECT project_id // This finds all projects worked
	//on by employee 301.(0.5 Mark)
	FROM works_on
	WHERE emp_id = 301
	);			

Q16.	Given student database: **STUDENT**(student_id, student_name, dept, city) **COURSE**(course_id, course_name, dept, credits) **ENROLLMENT**(student_id, course_id, marks, semester) a) List the names of students who have scored above the overall average marks of all the students across all courses using a nested SQL subquery. b) If the student table contains 1,00,000 records. A query frequently searches students using: **SELECT * FROM student WHERE ID = 12345;** The query execution is slow. Write an appropriate SQL	**3**	**2**	**3**

	statement to improve the search performance, and explain how it helps. **Answer: a) 1.5M +b)1.5M** a) SELECT student_name FROM STUDENT WHERE student_id IN (SELECT student_id FROM ENROLLMENT WHERE marks > (SELECT AVG(marks) FROM ENROLLMENT)); b) **CREATE INDEX student_id_index ON student(student_id);** **Or** **CREATE INDEX student_id_index ON student(ID);** This statement creates an index on the ID/student_id attribute. The index improves performance by allowing the database to locate records directly without scanning the entire table. **Marks Split up (1+0.5)** – 1 mark for writing the appropriate SQL statement, 0.5 mark for explaining how creating index helps to improve search performance			
Q17.	Illustrate three-level data abstraction (three-schema architecture) with a neat diagram and state the purpose of each level. **Answer: diag:1M+exp:1M→2M**  Physical level: describes how a record (e.g., instructor) is stored. Logical level: describes data stored in database, and the relationships among the data.	**2**	**1**	**2**

	View level: application programs hide details of data types. Views can also hide information (such as an employee's salary) for security purposes.			
Q18.	Distinguish between Relation Schema and Relation Instance by mentioning atleast 4 differences. **Answer: 2*0.5→2M** **Relation Schema** It is the logical structure or design of a relation. It defines the relation name and its attributes. It does not change frequently. Example: STUDENT (Reg_No, Name, Branch) **Relation Instance** It is the actual data stored in the relation at a particular time. It consists of rows (tuples) containing attribute values. It changes frequently when data is updated. Example: (101, Ravi, CSE), (102, Sneha, ECE)	**2**	**1**	**3**
Q19.	Consider the employee database given below and give an expression in the relational algebra to express the following query: Find the ID of each employee who does not work for "ABC Corp". **employee** (EID, person name, street, city) **works** (EID, CID, salary) **company (**CID, company name, city) **Answer:** $\pi_{EID}(employee) - \pi_{EID}(works \bowtie \sigma_{company_name = "ABC\ Corp"}(company))$ Scheme: $\pi_{EID}(employee)$– 0.5mark $\pi_{EID}(works \bowtie \sigma_{company_{name}} = \text{ABC Corp}(company) - 1\ mark$ Minus operator – 0.5 Mark	**2**	**1**	**3**